

MS5612: Real Options Valuation

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Contents

1	27th July - Monday	3
1.1	DCF vs Options Approach	3
2	29th July - Wednesday	3
2.1	NPV	4
2.2	Class Discussion on Strategic Investment	4
2.3	Example	4
2.3.1	Case 1: Cost Certain, Revenue uncertain	4
2.3.2	Case 2: Revenue certain at 90, cost uncertain	4
2.3.3	Case 3: Revenue and cost are uncertain	5
3	03rd Aug - Monday	5
3.1	Summary of the last class	5
3.2	Options	5
3.3	Options Payoff	6
3.4	Popular Combinations of Options	6
3.5	Let's get a little more creative	6
3.6	Some real world examples of options	7
3.7	Course Project	7

1 27th July - Monday

Finance could broadly be divided into 3 parts: 1. Personal Finance, Corporate Finance, and Financial Markets

How do businesses manage their financing? - Corp Fin is mostly about it.

Corp Fin could again broadly be classified into two: Investment and Financing.

Investment decisions result in assets and financing decisions result in liabilities.

Stocks could be valued in any of the following methods:

- Comparative method
- Dividend Discount method
- Free Cash Flow method
- Book Value

How do we value Debt?

- present value of cashflows of the debt (interest + principal)

How do we value projects (assets)?

- Discounted Cashflow Method (DCM)

Keyword: Fortune 500 Keyword: Managers **create** shareholder value.

Most Fortune 500 firms use real options. But as per McKinzie's India head, almost no Indian company is using real options. So, there is an opportunity.

1.1 DCF vs Options Approach

- DCF involves limited discretionary power, but options provide flexibility.
- If people are liking flexibility, we may value this flexibility.
- Managers create flexibility where seemingly none exists.

2 29th July - Wednesday

Focus of the course is on the management of long term assets (not much on liabilities).

NPV, IRR, Profitability Index (PI), Payback Period.

NPV - How much does the project worth? (**How much?**)

$$NPV = \sum_{i=1}^n \frac{C_i}{(1+r)^i} - IO$$

Keyword: IO - Initial Outlay

IRR - What returns are we making? (**What percent?**)

PI - Present value of cash inflow/ initial outlay (IO) (**What rank?**)

PB - **How long?**

2.1 NPV

$$NPV = \sum_{i=1}^n \frac{C_i}{(1+r)^i} - IO$$

Investment rule: if $NPV > 0$ we may accept
if $NPV < 0$ we must reject
if $NPV = 0$ we reject it as it doesn't create any shareholder value.

2.2 Class Discussion on Strategic Investment

What are Strategic Investments?

How do you justify an Investment? NPV, IRR, PB, PI? But only these, really?

Some investments could have negative NPV, but they could still be chosen.

Strategic investments are those that may fail the test of conventional investment tests (NPV, IRR) but they may pass the tests in different cases.

2.3 Example

Let's say there's an RnD project requiring 15 million USD.

To manufacture the plant, the cost could be 40 million, 80 million, and 120 million.

And the market share could be 50 million, or 130 million

The expected cost is 80 million, expected benefit is 90 million, and 15 millions for RnD.

Here, conventionally, the recommendation is to not invest and the NPV is -5 million USD.

2.3.1 Case 1: Cost Certain, Revenue uncertain

Let's say once we've invested in RnD, the manufacturing cost is certainly 80 million. If, now the market is 50 million, we do not invest. If however the market is 130 million, we go ahead and invest, in which case the benefit is 50 million. NPV in this case is $\frac{1}{2}(0 + 15) - 15 = 10$ million.

2.3.2 Case 2: Revenue certain at 90, cost uncertain

if cost is 40 mil, we invest, benefit = 50 mil
if cost is 80 mil, we invest, benefit = 10 mil
if cost is 120 mil, we do not invest, benefit = 0

$$NPV = \frac{1}{3}(50 + 10 + 0) - 15 = 5 \text{ million.}$$

2.3.3 Case 3: Revenue and cost are uncertain

$(40, 50) = 10$ $(40, 130) = 90$ $(80, 50) = 0$ $(80, 130) = 50$ $(120, 50) = 0$ $(120, 130) = 10$

$NPV = 1/6(10 + 90 + 50 + 10) - 15 = 11.67$ million

In the formula of NPV, r is the measure of risk.

The discount rate for a project must reflect the risk of the project.

Case 3 is definitely the riskiest, but what about the other cases? Which one is more risky? We can calculate the risk? We may also calculate the variance. We'll notice that the NPV increases as the risk increases.

So, this RnD investment is an option that would give the firm the flexibility to go ahead and set up a manufacturing plant. And the riskier the option is, the more valuable it is.

3 03rd Aug - Monday

3.1 Summary of the last class

- Corporates have to only invest in projects having NPV positive, else they won't be creating shareholder values.
- But often there are Strategic investments.
- SIs may have negative NPV using the conventional DCF method.
- Strategic investments are those that have several options and if we add the value of these options, we may have positive NPV.
- As the uncertainty increases the value of the initial investments is higher.

3.2 Options

- Options give the holder the right but not the obligation to do something.
- **Call Option:** Provides the right to buy the underlying
- **Put Option:** Provides the right to sell the underlying
- These are called derivatives because they derive their value from the underlying.
- **Option Value:** Premium paid to buy the option/ market price of the option.
- **Option Holder:** Person/entity who's the buyer of the option.
- **Option Seller:** Person/entity who's selling the option.

3.3 Options Payoff

- Stock - Simply a straight line with slope 1 against the value of Stock
- Risk free Bond - Simply a horizontal line against the value of Stock
- Call Option

For the buyer - $y = 0$ until $x = \text{strike}$, then a straight line slope 1. We may have to shift the graph downward by the amount paid as premium to get the profit.

For the Seller - reflection of what it is for the buyer along the x-axis.
- The upside to the seller is limited but the downside is unlimited. Vice versa for the buyer.
- Options are a zero-sum game. Someone's gain is someone's loss.
- Put Option

For the buyer - Reflection along $x = \text{strike}$ of the above curves

For the seller
- Limited upsides and limited downsides
- Call and Put options are vanilla options. The fun comes when we combine some options.

3.4 Popular Combinations of Options

- **Stock + Sold a Call:** Do from the graph
- **Risk Free Bond + Sold a Put:** Do from the graph
- The payoff of both the above are same.
- We do this exercise and can build a basket of options/securities to get the desired payoff.

3.5 Let's get a little more creative

- **Buying a Call + Buying a Put:** Almost a Kite, we make money, with high chance and the more volatile the stock is, the more chance.
- If we factor in the premium, there is a part on the x-axis for which the profit would be negative.
- **Long Call @ 100 + Short 2 Calls @ 110 + Long Call @ 120:** Butterfly Combination
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3.6 Some real world examples of options

- Uber rides cost a little higher for reservation rides.
- Indigo charges higher for the option of low price cancellation. (Indigo Fare vs Indigo UpFront)
- What are the factors that influence this premium charge for cancellation?
- The premium seems to be the same across flights, accounting for flight distance, and time.
- Cancelling is a Put option.
- These are the examples of real options.
- Is insurance a put option?

3.7 Course Project

- Step 1: Propose 3 three topics that we'd like to study
- Write a 100 words description on all three topics, indicating the use of real options.
- Submit hard copy by 5 pm, Aug 7th.
- Primary data and not desk based research.